

Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

✉ amirgolpaa24@gmail.com | ✉ amir.mirzaigolpayega@ucalgary.ca | ☎ +1 (587) 439-0436

🌐 [linkedin.com/in/amir-golpayegani-218148165](https://www.linkedin.com/in/amir-golpayegani-218148165) | 🐙 github.com/amirgolpaa24 | 🌐 Portfolio

Summary

M.Sc. Computer Science graduate with experience building Python/C++ systems for machine learning, predictive modeling, NLP-related backend services, large-scale data processing, and AI workflow automation. Skilled in PyTorch, pandas, NumPy, SQL, model evaluation, analytical pipelines, and translating complex datasets into practical tools for decision support.

Education

M.Sc. in Computer Science <i>University of Calgary</i>	GPA: 3.94/4.0	Sep 2022 – Feb 2026 <i>Calgary, Canada</i>
B.Sc. in Computer Engineering <i>Sharif University of Technology</i>	GPA: 3.69/4.0	Sep 2016 – Feb 2021 <i>Tehran, Iran</i>

Technical Skills

Programming: Python, C++, JavaScript, SQL
Data Science / ML: PyTorch, Scikit-learn, TensorFlow, CNNs, model evaluation, feature engineering, predictive analysis
Data / Analytics: NumPy, Pandas, SciPy, PostgreSQL, data pipelines, preprocessing, statistical analysis, storage/retrieval workflows
NLP / AI: NLP service APIs, LLM workflows, AI agents, prompt design, n8n, OpenClaw, SearXNG, Telegram API
Engineering: Django, Node.js, REST APIs, Docker, Git, Linux, AWS, GitHub Actions, Streamlit

Work Experience

Research Software Developer Sep 2022 – Feb 2026
BigGeo / Mitacs, University of Calgary *Calgary, Canada*

- Built C++/Python software for scalable processing, storage, retrieval, and analysis of large scientific datasets.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGs-based encoding pipelines to convert heterogeneous imagery into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Documented data models, experimental assumptions, and retrieval results for research and stakeholder communication.

Back-End Developer Jun 2020 – Jul 2022
Asr Gooyesh Pardaz *Tehran, Iran*

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

Deep Learning Teaching Assistant Jul 2026
Neuromatch Academy *Remote, United States*

- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Teaching Assistant Dec 2022 – Apr 2025
Department of Computer Science, University of Calgary *Calgary, Canada*

- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** Assisted +75 students with SQL, database design, course projects, grading, and exam support.

Research and Technical Projects

DEM Super-Resolution Framework May 2024 – Feb 2026
Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

- Trained deep ResNet models for $4\times$ DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess

mountainous terrain data worldwide.

- Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
- Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.

Spatiotemporal Earth Observation Data System Using DGGS

Sep 2022 – Feb 2026

[First-author publication](#) | [M.Sc. thesis](#) | C++, Python, DGGS, Wavelets, B-splines, Time-Series

- Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Implemented retrieval workflows for structured spatiotemporal datasets and evaluated query behavior.
- Retrieved 105,489 DGGS cells per time-series query in 13.9s, compared with about 220s using global retrieval.

Agentic Job Search Automation Platform

May 2026 – Present

OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API

- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, application decision support, and structured output generation.
- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.

Persian Text Normalizer for NLP

Oct 2016 – Feb 2021

Python, NLP, Text Processing

- Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines.
- Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGS*. Remote Sensing, 2025. [doi:10.3390/rs17040570](#)

Certificates

Supervised Machine Learning: Regression and Classification,

DeepLearning.AI and Stanford University, Coursera

Oct 2024

Applied Machine Learning in Python,

University of Michigan, Coursera

Jul 2020

Languages

English: Full Professional Proficiency (TOEFL 115/120)

Persian: Native Proficiency

Achievement

Teaching Assistant Excellence Laureate

2024/2025

Department of Computer Science, University of Calgary — awarded for outstanding teaching support ([link](#))

Ranked 17th Nationwide in Iran University Entrance Exam ([Konkour](#))

May 2016

Top 0.001% among all participants